

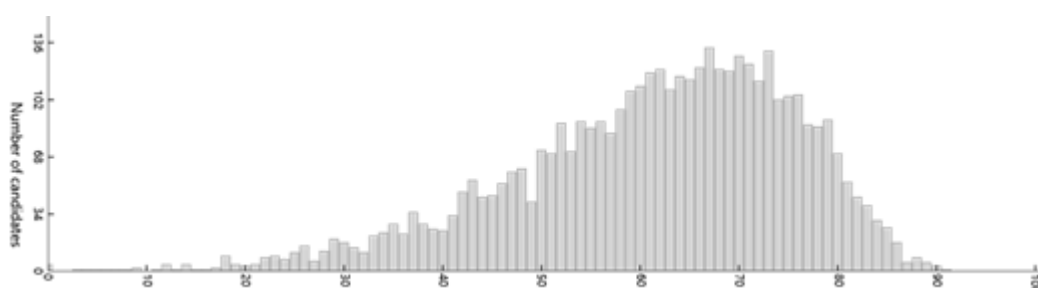


2021 ATAR course examination report: Human Biology

Year	Number who sat	Number of absentees
2021	4216	78
2020	4380	64
2019	4475	50
2018	4962	54

The number of candidates sitting and the number attempting each section of the examination can differ as a result of non-attempts across sections of the examination.

Examination score distribution–Written



Summary

Attempted by 4216 candidates Mean 61.26% Max 90.77% Min 0.00%

The examination discriminated well, producing scores from 0.00% to 90.77%. The examination was an appropriate length with most candidates attempting all questions. The overall mean of 61.26% was a solid improvement on the 55.61% mean attained in 2020.

Section means were:

Section One: Multiple-choice	Mean 77.06%		
Attempted by 4216 candidates	Mean 23.12(/30)	Max 30.00	Min 0.00
Section Two: Short answer	Mean 56.50%		
Attempted by 4205 candidates	Mean 28.25(/50)	Max 45.75	Min 0.00
Section Three: Extended answer Unit 3	Mean 59.04%		
Attempted by 4152 candidates	Mean 5.90(/10)	Max 10.00	Min 0.00
Section Three: Extended answer Unit 4	Mean 39.88%		
Attempted by 4132 candidates	Mean 3.99(/10)	Max 9.75	Min 0.00

General comments

As indicated by the mean of 61.26%, candidates were able to demonstrate a sound knowledge of the Human Biology syllabus throughout the paper. Section One: Multiple-choice was particularly strong with very few questions proving to be a challenge to the majority of candidates. In both Sections Two and Three, candidates could recall basic facts, but only the top candidates could effectively answer the in-depth analysis questions.

Advice for candidates

- The examination is based on the syllabus and not a textbook. Candidates should expect a comprehensive coverage of the syllabus points in the examination; however, not every

syllabus point will be examined. Although the same syllabus points may appear in consecutive examinations, questions will be structured in new and different contexts.

- Candidates must read questions carefully to ensure they understand the meaning of verbs used in the question. Candidates should refer to the *Glossary of key words used in the formulation of questions*, available on the course page, to aid them in knowing the difference between a question requiring them to 'name' or 'identify' compared to one requiring them to 'describe' or 'explain'.
- Candidates are encouraged to answer every question. Candidates should always put something down as an answer, even if it is an educated guess. A non-attempted question means no marks can be achieved.
- Candidates need to focus on interpreting the question and providing a concise answer, rather than simply writing down all they have learnt on the topic. Stating key terminology or memorised facts will often not answer the entire question. Instead, candidates need to engage higher-order thinking skills and apply their knowledge.
- The most thorough and complete responses in the Extended answer section of the paper are often produced by candidates who have completed a plan. Candidates should take time to plan responses and ensure all parts of the question are addressed.
- Candidates are encouraged to present annotated diagrams, charts or tables to construct responses to questions in the short and extended answer sections. This technique is particularly important in the extended answer section to help candidates write clear and precise answers and ensure that markers can easily follow and award marks for responses.

Advice for teachers

- Candidates require a detailed knowledge of all syllabus points and the ability to apply this to new and different contexts. Simple recall and rote learning of facts are not enough for candidates to gain full marks in the examination. Candidates must be able to think critically and apply their knowledge to unique scenarios.
- There were several specific syllabus points that were not answered well in this examination. Teachers should be aware of these areas and design appropriate teaching strategies around them. These include:
 - o inflammation as a non-specific response
 - o theory of natural selection
 - o factors that produce new mutations
 - o linking comparative studies of anatomy to the construction of phylogenetic trees
 - o limitations in the fossil record.
- Teachers are reminded that textbooks are not the syllabus. Supplementary teaching is necessary and learning with other references and resource materials beyond the textbook is essential.
- Examination technique should be modelled and reinforced throughout the year during classroom lessons and assessments. Students need to be given ample practise at decoding extended response questions as part of their lessons and school-based assessment program. They need substantial practise in breaking down a question and identifying the question components.
- Mathematical skills are a required element in the syllabus. Students need ample practise at applying all required mathematical skills in a scientific method context.

Comments on specific sections and questions

Mean score for Section One questions ranged from 36.29% to 96.04%. Some questions were answered very successfully in this section and demonstrated an outstanding ability of factual recall of specific syllabus points such as osmoregulation, thermoregulation, immunity, adaptations to bipedalism and the divisions of the nervous system. Section Two questions had mean scores ranging from 50.47% to 68.52%. Questions relating to the structural comparisons between humans and great apes were completed the most successfully. The

Extended answer questions on Unit 3 were done very well compared to previous years, with a high overall mean score of 59.04%. Question 38 on homeostasis was the most successfully answered with a mean of 61.70%, while Question 39 had a lower mean of 55.60%. The Extended answer Unit 4 section was the most demanding section of the paper with the lowest overall mean of the four sections (39.88%). The mean scores of the Unit 4 questions were comparable at 41.85% for Question 40 and 38.54% for Question 41.

Section One: Multiple-choice (30 Marks)

This section had an overall mean of 77.06%. The maximum mark achieved in this section was 100% and the minimum mark was 0%. Generally, questions were well answered with only Questions 5, 6, 10, 22, 28 and 30 having mean scores at or below 60%. Questions 2, 4, 7, 14, 16, 17, 18, 19, 20, 21, 24, 25 and 27 were the easiest with mean scores all above 85%. Candidates performed well on straightforward recall style questions and found the more complex, multi-step questions challenging.

Question 1 attempted by 4216 candidates	Mean 0.69(/1)	Max 1	Min 0
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Question 2 attempted by 4216 candidates	Mean 0.92(/1)	Max 1	Min 0
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Question 3 attempted by 4216 candidates	Mean 0.78(/1)	Max 1	Min 0
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Question 4 attempted by 4216 candidates	Mean 0.86(/1)	Max 1	Min 0
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Question 5 attempted by 4216 candidates	Mean 0.51(/1)	Max 1	Min 0
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Candidates appeared to omit the role of the pituitary as one of the endocrine glands responsible for controlling metabolic rate.

Question 6 attempted by 4216 candidates	Mean 0.54(/1)	Max 1	Min 0
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The distractor, which incorrectly included proteins and RNA as part of comparative genomics, proved appealing to a significant number of candidates.

Question 7 attempted by 4216 candidates	Mean 0.86(/1)	Max 1	Min 0
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Question 8 attempted by 4216 candidates	Mean 0.69(/1)	Max 1	Min 0
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Question 9 attempted by 4216 candidates	Mean 0.80(/1)	Max 1	Min 0
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Question 10 attempted by 4216 candidates	Mean 0.57(/1)	Max 1	Min 0
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A fair proportion of candidates misinterpreted the example in the question and failed to distinguish the independent variable as the supplement and not the amount of calcium.

Question 11 attempted by 4216 candidates	Mean 0.70(/1)	Max 1	Min 0
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Question 12 attempted by 4216 candidates	Mean 0.84(/1)	Max 1	Min 0
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Question 13 attempted by 4216 candidates	Mean 0.80(/1)	Max 1	Min 0
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Question 14 attempted by 4216 candidates	Mean 0.96(/1)	Max 1	Min 0
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Question 15 attempted by 4216 candidates	Mean 0.71(/1)	Max 1	Min 0
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Question 16 attempted by 4216 candidates	Mean 0.87(/1)	Max 1	Min 0
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Question 17 attempted by 4216 candidates	Mean 0.92(/1)	Max 1	Min 0
Question 18 attempted by 4216 candidates	Mean 0.89(/1)	Max 1	Min 0
Question 19 attempted by 4216 candidates	Mean 0.86(/1)	Max 1	Min 0
Question 20 attempted by 4216 candidates	Mean 0.85(/1)	Max 1	Min 0
Question 21 attempted by 4216 candidates	Mean 0.93(/1)	Max 1	Min 0
Question 22 attempted by 4216 candidates	Mean 0.36(/1)	Max 1	Min 0
This question was the most challenging question. The question required a precise understanding of the terminology used to explain natural selection.			
Question 23 attempted by 4216 candidates	Mean 0.77(/1)	Max 1	Min 0
Question 24 attempted by 4216 candidates	Mean 0.94(/1)	Max 1	Min 0
Question 25 attempted by 4216 candidates	Mean 0.94(/1)	Max 1	Min 0
Question 26 attempted by 4216 candidates	Mean 0.82(/1)	Max 1	Min 0
Question 27 attempted by 4216 candidates	Mean 0.86(/1)	Max 1	Min 0
Question 28 attempted by 4216 candidates	Mean 0.56(/1)	Max 1	Min 0
Candidates had difficulty distinguishing between the alpha and beta forms of Thalassemia in this question.			
Question 29 attempted by 4216 candidates	Mean 0.73(/1)	Max 1	Min 0
Question 30 attempted by 4216 candidates	Mean 0.59(/1)	Max 1	Min 0
This question addressed the ethical considerations of animal research and showed that a significant number of candidates incorrectly believe financial cost is an ethical consideration.			

Section Two: Short answer (106 Marks)

This section had an overall mean of 56.50%. The maximum mark achieved on the section was 91% and the minimum mark was 0%. Questions 33, 35 and 37 all had mean scores above 55%. Question 33 focused on structural comparisons between humans and great apes and adaptations to bipedalism and was the most successfully completed question in this section. Questions 31, 32, 34 and 36 all had very similar mean scores, ranging from 50.47% to 52.85%. Question 36 on comparisons between nervous and hormonal systems, blood glucose regulation, diabetes and recombinant DNA technology was the most challenging question of Section Two. Throughout the section, candidates were able to demonstrate a good recall of basic facts and knowledge. The higher-order and application questions were successful in differentiating candidates and provided an opportunity for candidates to demonstrate their understanding of course content.

Question 31 attempted by 4198 candidates Mean 6.20(/12) Max 12 Min 0
Candidates performed well on the structure and function recall style questions in parts (a), (b)(i) and (c)(i) of the question. Parts not done well in the question included parts (b)(ii), (c)(ii) and (d). All of these required higher-order thinking and explanation of how the nervous system operates. In part (b)(ii), candidates often could not link the presence of grey matter

being due to neurons lacking myelin and the cell bodies of neurons being located in that region. Part (c)(ii) appeared to present a problem with question interpretation, with candidates often failing to describe both the detection of the sensation and the pathway required for it to be registered, as was required by the question. Candidates struggled in part (d) to communicate where the damage would be located in the spinal cord, with many incorrectly discussing motor pathways and lacking precise terminology in their responses.

Question 32 attempted by 4193 candidates Mean 9.00(/17) Max 17 Min 0
The scientific method parts of the question were completed well, with candidates able to write an effective hypothesis and identify the purpose of the control group. In part (a), candidates were generally able to discern the type of mutation as being a germline one, but in part (b), their explanations on why it was that type of mutation was lacking the precise wording required to explain their choice. It was evident in part (c) that candidates had a good basic understanding of the causes of the mutations, however many responses lacked scientific terminology. A significant number of candidates stated that mutagens 'increase mutations', this was already stated in the question and also showed no insight into the concept. Candidates needed to be able refer to mutations as permanent changes in the DNA and how these changes occur, in order to achieve marks on this question. In part (e), candidates needed to refer to the changes in the allele frequencies with the correct terminology. Responses often incorrectly interchanged the term gene and allele. For full marks on the question candidates needed to correctly refer to the impact of the new mutated allele on the gene pool.

Question 33 attempted by 4203 candidates Mean 7.54(/11) Max 11 Min 0
This question was the most successfully completed in this section. All parts of the question had good mean scores and candidates demonstrated a sound understanding of the structural comparisons between humans and great apes and adaptations to bipedalism. It should be noted however, that in part (c) candidates that scored lower marks on the question often did so because of their lack of precision in using appropriate terminology to describe the structural features. Incorrect responses included vaguely describing that 'foramen magnum is at the back'; not indicating that the foramen magnum is on the skull or indicating it was part of the spine; mixing up the meaning of opposable and non-opposable; and mentioning thumb length but not in comparison to finger length.

Question 34 attempted by 4168 candidates Mean 6.11(/12) Max 12 Min 0
All parts of the question focusing on Tay Sachs were answered relatively poorly, reflecting a limited understanding of this disease and the impacts of it on gene pools. Even part (a) of the question, which only required a general statement about Tay Sachs, had a mean score below 50%. The application of Tay Sachs to the founder effect presented problems. The response to the question followed the well-established standard four marks for the four major components required in the process of the founder effect, yet a significant number of candidates could not produce that response. In part (c), concentrating on PCR was done well but it should be noted that many candidates only mentioned primers during the annealing stage and did not mention that the primer is attached to the complementary bases. For part (d), candidates often omitted the key elements of variation and isolation in the natural selection process and only focused on the survival advantage component.

Question 35 attempted by 4198 candidates Mean 12.32(/21) Max 21 Min 0
This question produced a mix of well answered and poorly answered parts. Recall style questions produced higher mean scores while questions requiring more in-depth explanations proved more difficult. Part (d) produced a pleasing mean with candidates able

to effectively distinguish antibiotics and vaccines. Candidates could correctly identify vector transmission but often failed to clearly explain person to person disease transmission in part (b)(ii). Answers to this part often lacked precise descriptions of where the pathogen is hosted, how it is transmitted and how it is introduced into the body of the new host. Part (c)(i) showed that few candidates could identify that inflammation is a non-specific response and many incorrectly included heparin as a key part of the process. The wording in part (e) stated that candidates needed to identify reasons for why people participate in immunisation programs. However, many candidates incorrectly focused on reasons why people do not participate.

Question 36 attempted by 4193 candidates Mean 9.59(/19) Max 19 Min 0

This question was the least successfully completed of Section Two. It focused on comparisons between nervous and hormonal systems, blood glucose regulation, diabetes and recombinant DNA technology. Parts of the question were completed well, including part (a) identifying the neurotransmitter, part (c)(i) identifying the faster pathway and part (d)(i) identifying the hormone. However, the remaining parts of the question had a large number of candidates providing less than satisfactory responses. One significant incorrect response that appeared far too often in part (b) was that a neurotransmitter travels 'along a neuron'. The in-depth responses to explain the effect of diabetes and recombinant DNA technology proved challenging for the average candidate with only some candidates able to provide comprehensive responses. Part (c)(ii) averaged the lowest mark of the whole examination. Candidates appeared to misinterpret the question with many responses focused on the role of the myelin sheath in speeding up electrical impulses when the question actually required them to contrast the nervous and hormonal pathways.

Question 37 attempted by 4197 candidates Mean 9.12(/14) Max 14 Min 0

All parts of the question were done generally well with mean scores above 50%. Part (a), while looking simple on the surface, proved challenging with numerous candidates unable to name the species. Incorrect responses often included naming species not studied within the syllabus or incorrectly formatting the name following the genus and species nomenclature rules. In part (d)(iii), most candidates could correctly identify the skull as *Homo neanderthalensis* but very few were able to provide a valid reason as to why this species had a larger cranial capacity than modern humans.

Section Three: Extended answer Unit 3 (20 Marks)

This section had an overall mean of 59.04%. The maximum mark achieved on the section was 100% and the minimum mark was 0%. Question 38 was the most successfully answered question with a mean of 61.70%, while Question 39 achieved a mean of 55.60%. Generally, candidates who attempted questions using annotated diagrams and tables were able to construct more precise and understandable responses.

Question 38 attempted by 2314 candidates Mean 12.34(/20) Max 20 Min 0

This question was favoured by candidates over Question 39 and was the most successfully completed extended answer of all four options in the examination. Part (a) on body temperature regulation was done particularly well with very detailed and complete responses from a large number of candidates. Those who constructed a feedback loop diagram were the easiest to interpret and mark. In part (b), candidates often did not make the link as to why body fluid regulation was required in the situation, but generally provided very good answers on the components of the feedback loop.

Question 39 attempted by 1838 candidates Mean 11.12(/20) Max 20 Min 0

While not as popular as Question 38, this question still produced a pleasing overall mean. Part (a) was the more challenging part of the question. While candidates often seemed to know the basics of the pathway of the spinal reflex, answers often lacked exactitude, missing the key elements of the dorsal and ventral root, location of grey matter, position of synapses and the end point connection to a muscle. Part (b) was well-attempted, in particular responses on the transmission across the synapse was effectively communicated.

Section Three: Extended answer Unit 4 (20 Marks)

This section had an overall mean of 39.88%. The maximum mark achieved on the section was 97.5% and the minimum mark was 0%. Questions 40 and 41 had comparable mean scores of 41.85% and 38.54% respectively. Generally, candidates who attempted to answer the questions using annotated diagrams and tables were able to construct more precise and understandable responses.

Question 40 attempted by 1634 candidates Mean 8.37(/20) Max 19.5 Min 0

This question was the least attempted question of the two options in Unit 4. In part (a), many candidates incorrectly focused on potassium-argon dating as a possible technique. The marking key also separated the principle of superposition, index fossils and correlation of rock strata into separate marking blocks but candidates often responded by putting all three processes together. While this did not penalise the candidates it did make the marking of the response more difficult. Part (c) showed limited achievement, with candidates struggling to provide eight valid points associated with mitochondrial DNA.

Question 41 attempted by 2498 candidates Mean 7.71(/20) Max 17.5 Min 0

This question was popular amongst candidates but recorded the lowest overall mean score; candidates were challenged by all parts of this question. In part (a), candidates often superficially explored the problems with the fossil record and lacked precise, detailed explanations of the issues. A common misuse of terminology was seen with candidates referring to the conditions the 'fossil needed for fossilisation' instead of the conditions the 'organism needed for fossilisation'. Part (b) was frequently misinterpreted by candidates as only requiring an explanation of the comparative studies of anatomy. However, the question required the linkage of the comparative studies to the construction of phylogenetic trees and many candidates simply did not address this element of the question.